

// research roadmap · 12 papers · 6 verification domains

Proof-of-Control. Research Roadmap

A prioritized research agenda following from *Proof-of-Control: An Open Standard for Runtime Verifiability and Cryptographic Oversight in Autonomous AI Execution*. Writing the standard, building the reference implementation, and putting the paper through review surfaced questions the specification currently answers by assertion, by convention, or not at all. This document turns those into a research program.

12

SCOPED PAPERS

127

REQUIREMENTS IN
THE STANDARD

6+4

DOMAINS + CROSS-
CUTTING CHAPTERS

3

CORRECTIVE — THE
STANDARD IS
WRONG WITHOUT
THEM

If you only do three, do these.

Each paper is scoped so it can be handed to a deep-research pass and come back with something worth building on: a specific question, the prior art it must engage, the experiment that would settle it, and the change it would make to the standard either way.

P01 A Trust Calculus for Attestation Tiers

C8 VERIFIABILITY TIERS · CORRECTIVE + FOUNDATIONAL · IMPACT: VERY HIGH

The standard's central taxonomy is provably wrong as written — found with a hardware measurement, not an argument. Which parties must a verifier still trust, and can that residual set be *computed* from a system description rather than argued about?

P02 From Message Authorization to Effect Binding

C4 AUTHORIZATION / C7 EVIDENCE · CORRECTIVE + FOUNDATIONAL · IMPACT: VERY HIGH

The paper's main theorem proves less than the paper claims, and everything downstream inherits the gap. A signed capability binds the *bytes of a request* — under what conditions does it bind the *effect a real endpoint performs*?

P05 Accountable but Unlinkable Agent Identity

C5 IDENTITY · GENERATIVE · IMPACT: HIGH · EFFORT: LARGE

The largest genuinely open design question in the standard, with mature cryptography waiting to be applied: verifiable and attributable *when it matters*, unlinkable across relying parties the rest of the time.

WHY THESE — P01 and P02 are *corrective*: the standard overclaims without them. P05 is *generative*: it opens a design space rather than closing a hole.

WHERE THIS COMES FROM — NOTHING HERE IS INVENTED

- **Open working-group issues.** Appendix D records 13, several contributed by working-group members.
- **Threat-model gaps.** Appendix C grades 32 threats; most are *Partial*.
- **Review findings.** Three independent reviews returned weak reject with blocking objections; two unaddressed objections become P01 and P02.
- **Measurements that surprised us.** The TDX deployment showed a hardware quote costs 39.5 ms — forcing a new requirement, and P03.
- **Defects found by building.** Three requirements exist because the implementation was wrong in a way the prose could not reveal. Every paper here has something to build, not only something to argue.

All twelve papers, by priority.

Corrective — the standard currently says something untrue or unproven without it. **Foundational** — other work depends on it. **Generative** — it opens design space. **Empirical** — it measures something asserted but unmeasured.

#	PAPER	DOMAIN	KIND	IMPACT	EFFORT
P01	A Trust Calculus for Attestation Tiers Which parties must a verifier still trust — and can the residual set be computed?	C8 Tiers	Corrective · Foundational	VERY HIGH	Medium
P02	From Message Authorization to Effect Binding When does binding the request bind the effect the endpoint performs?	C4 / C7	Corrective · Foundational	VERY HIGH	Medium
P03	The Cost of Knowing What Ran Attestation refresh policy, and what is exposed between quotes.	C7 Evidence	Empirical · Corrective	HIGH	Small
P04	Are Bounded Path Summaries Soundly Bounded? A bounded summary of unbounded history is lossy — can an adversary exploit the loss?	C4 Authorization	Corrective · Foundational	HIGH	Small
P05	Accountable but Unlinkable Agent Identity Attributable when it matters, unlinkable the rest of the time — at what cost?	C5 Identity	Generative	HIGH	Large
P06	Delegation Without Amplification A delegate's authority must be a strict subset — and chains must not reassemble authority nobody held.	C4 / C5	Generative · Foundational	HIGH	Medium
P07	Evidence Continuity Across Trust Domains What construction makes the seam between attestation domains verifiable?	C3 Portability	Generative	MED-HIGH	Medium
P08	What a Serial Log Costs a Fleet A hash chain is serial; what does that cost concurrent agents?	C7 Evidence	Empirical	MED-HIGH	Small
P09	Evidence That Reveals Nothing Prove actions satisfied policy without revealing actions, policy, or data.	C2 Privacy	Generative	MEDIUM	Large
P10	What Belongs in an Agent Bill of Materials An agent's effective composition changes continuously — what does an AgBOM record?	C1 Provenance	Generative	MEDIUM	Medium
P11	Who Pays for Verification The economics and incentives that decide whether any of this is adopted.	Cross-cutting	Generative	MEDIUM	Medium
P12	Why Assurance Regimes Fail What the history of assurance standards predicts about this one.	C10 Conformance	Empirical	MEDIUM	Medium

Six verification domains, and what each still needs.

C1 Provenance

13 reqs

Solid mechanisms (signing, SBOM), weak on *runtime* composition.

P10

C2 Privacy

13 reqs

Specified a zero-knowledge path and never built it.

P09

C3 Portability

5 reqs

Thinnest domain; the cross-boundary evidence chain is genuinely unsolved.

P07

C4 Authorization

12 reqs

Best-developed, but the central theorem is weaker than claimed.

P02

P04

P06

C5 Identity

6 reqs

Underspecified — and the open questions are the most interesting in the standard.

P05

P06

C6 Security

12 reqs

Leans hardest on hardware, which is where the trust story breaks.

P01

P03

C7–C10 Cross-cutting chapters

66 reqs

C7 evidence is the deepest chapter; C8 verifiability tiers are the weakest link — the tier definitions are where the standard overclaims most.

P01

P03

P08

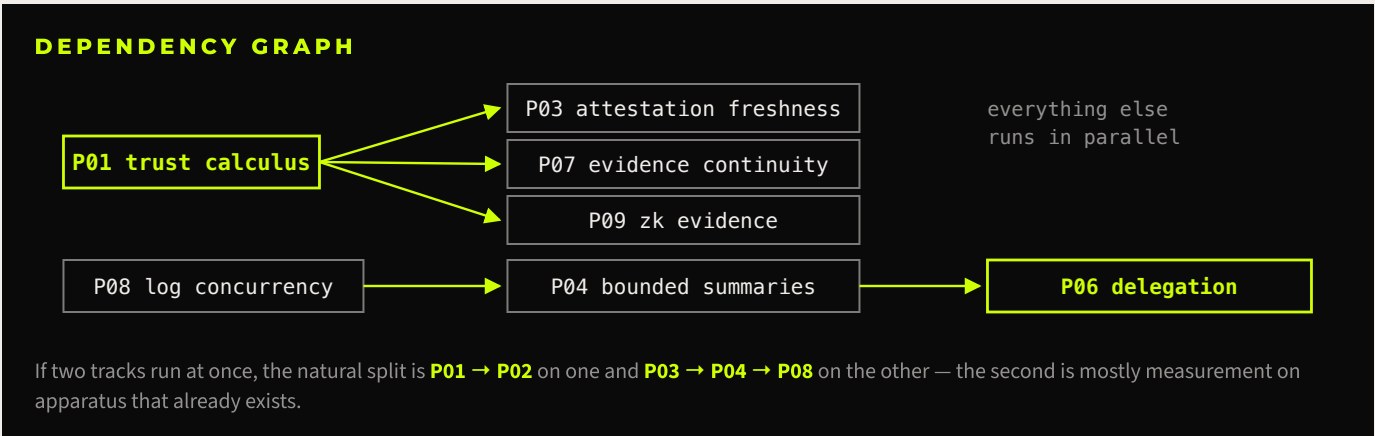
P11

P12

Two observations worth noting. C3 Portability has five requirements — the least developed domain and the one where an evidence chain most obviously breaks, which is a poor combination. And C5 Identity has six requirements but carries two of the three open issues the working group flagged: the specification is thin exactly where the design questions are hardest.

The order, and the dependencies that bind.

DO FIRST	P01 · P02	Corrective and foundational. The standard is wrong without them.
DO CHEAPLY, SOON	P03 · P04 · P08	Small effort, existing apparatus, and P04 corrects a published claim.
THE REAL RESEARCH	P05 · P06 · P07	Genuine open design space; where the interesting results are.
LONGER HORIZON	P09 · P10	Large effort, or dependent on the foundational work.
INSTITUTIONAL	P11 · P12	Not technical — and possibly more determinative of adoption than any of the above.



WHERE THIS IS MOST LIKELY WRONG

P11 and P12 may be underrated. Certificate Transparency succeeded because browsers mandated it, not because the Merkle tree was elegant. If that pattern holds, P11 is the most important paper here — and it is ranked tenth.	P05 may be overrated per effort. The measurement half — what anonymous credentials cost at agent action rates — is small, decisive, and should be done first regardless.
P04 may be too small to be a paper. Ranked fourth on importance, not publishability; those come apart here.	The corrective bias. Three of the top four fix our own errors. Someone without our history would rank P05 and P06 higher.

Anatomy of a paper brief.

Each paper file is written to be handed directly to a deep-research pass. Start the pass from sections 3 and 4; section 5 is the part that needs a human.

01 The question

One sentence, falsifiable.

02 Why it matters

Impact on the standard and on the field, stated separately.

03 What is already known

The literature to engage, with specific anchors.

04 What is genuinely open

The part nobody has answered.

05 Method

What to build and what to measure.

06 What would settle it

The result that would change minds — in both directions.

07 Consequence for the standard

Which requirements change, and how.

08 Venue and effort

Where it publishes and what it costs.

DELIBERATELY NOT IN THIS ROADMAP

- × **Better agent benchmarks.** Real and needed, but a different research programme.
- × **Model-level safety.** Outside the standard's determinism boundary — it verifies execution facts, not output quality.
- × **Hardware attacks on TEEs.** An active field with its own community; we consume its results.
- × **Formal verification of the whole standard.** Premature while the tier definitions are still wrong.

This is a planning repository.

No paper here has been written. Priorities reflect one assessment and should be argued with. Proof-of-Control is stewarded by the Advanced AI Society.

join a working group → advancedaisociety.org